Merge & Quick Sort:

31 58 10 23 30 49 50 24 22 51

Merge Sort:

It recursively divides the array into halves until sub-arrays of size 1 are reached, then merges them in sorted order.

Divide:

31 58 10 23 30 | 49 50 24 22 51

[31 58] [10 23 30] | [49 50] [24 22 51]

[31] [58] [10] [23] [30] | [49] [50] [24] [22] [51]

Merge:

[31 58] [23 30] | [49 50] [22 51]

[10] [23 30] ⇒ [10 23 30] | [24] [22 51] = [22 24 51]

[31 58] [10 23 30] = [10 23 30 31 58] | [49 50] [22 24 51]

⇒ [22 24 49 50 51]

Final merge

[10 22 23 24 30 31 49 50 51 58]

Quick Sort:

Quick sort uses a pivot element to partition the array into sub-arrays: elements less than the pivot and elements greater than the pivot. we will use the last element as the pivot for each step.

i) iter 1:

Array [31 58 10 23 30 49 50 24 22 51]

Pivot : 51

Partition: [31 10 23 30 49 50 24 22], 51, [58]

ii) iter 2:

Sub array: [31 10 23 30 49 50 24 22]

Pivot : 22

Partition : {10}, 22, {31, 23, 30, 49, 50, 24}

iii) iteration 3:

Sub array: [31 23 30 49 50 24]

Pivot : 24

Partition: {23}, 24, {31 30 49 50}

iv) iter 4:

Sub array: [31 30 49 50]

pivot : 50

Partition : {31 30 49} 50 [ ]

v) iter 5:

Sub array: [31 30 49]

Pivot : 49

Partition: {31 30}, 49, [ ]

vi) Iter 6:

Sub array: [31 30]

Pivot : 30

Partition : { } 30 {31}

now by combining Partition:

=[10 22 23 24 30 31 49 50 51 5